MILITARY SPECIFICATION

SODIUM CHLORATE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade sodium chlorate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Sodium chlorate shall be colorless or white crystals.

3.2 Chemical characteristics. Sodium chlorate shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as $NaClO_3$)	99.0	-----	4.2.4.1
Insoluble matter	----	0.010	4.2.4.2
Bromate (BrO_3)	----	0.07	4.2.4.3
Chloride (Cl)	----	0.005	4.2.4.4
Nitrogen compounds (as N)	----	0.001	4.2.4.5
Sulfate (SO_4)	----	0.003	4.2.4.6
Calcium, magnesium, and ammonium hydroxide precipitate	----	0.01	4.2.4.7
Heavy metals (as Pb)	----	0.001	4.2.4.8
Iron (Fe)	----	0.0005	4.2.4.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium chlorate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/2 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/2 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as $NaClO_3$).

(a) Acid ferrous sulfate solution. Dissolve 7.00 grams (g) of clear ferrous sulfate crystals in 90 milliliters (ml) of water which has been freshly boiled and then cooled to room temperature. Add 96-percent sulfuric acid to make 100 ml of solution. This solution should be prepared daily.

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(b) Procedure. Dry a small quantity of the specimen over sulfuric acid for 3 hours. Weigh to nearest milligram (mg) approximately 0.1 g of the dried specimen and dissolve it in 10 ml of water in a 250-ml flask. Add 35.0 ml of acid ferrous sulfate solution prepared as specified in (a), stopper the flask with a valve stopper to prevent access of air, and boil for 10 minutes. Cool to room temperature, add 10 ml of 10-percent manganous sulfate solution, and titrate the excess acid ferrous sulfate solution with 0.1N potassium permanganate solution. Run a blank in the same manner as the specimen test. Calculate the percent sodium chlorate as follows:

$$\text{Percent sodium chlorate} = \frac{1.774\ (A-B)\ C}{W}$$

where: A = Milliliters of potassium permanganate solution used to titrate the blank,
B = Milliliters of potassium permanganate solution used to titrate the specimen,
C = Normality of potassium permanganate solution used, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest 0.1 g approximately 20 g of the specimen and dissolve in 100 ml of hot water. Heat on a steam bath for 1 hour. Filter any insoluble residue through a tared filter crucible weighed to the nearest 0.1 mg, wash with hot water, and dry for 1 hour at 105° to 110° C. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A-B)}{W}$$

where: A = Weight of filter crucible and residue in grams,
B = Weight of filter crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Bromate. Dissolve 2.00 g of the specimen in 150 ml of water which has been freshly boiled and then cooled to room temperature in a glass-stoppered flask. Add 10.0 ml of 1.0N hydrochloric acid, 10 ml of freshly prepared 10-percent potassium iodide solution, and 5 ml of freshly prepared starch solution. Stopper immediately, allow to stand for 1 hour protected from light, and titrate with 0.01N sodium thiosulfate solution. No more than 6.5 ml of the sodium thiosulfate solution shall be required to reach the end point.

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4.2.4.4 Chloride. Dissolve 2.000 g of the specimen in 40 ml of water, cool to room temperature, add 5 drops of nitric acid free from oxides of nitrogen and 2 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced when 0.10 mg of chloride ion is substituted for the specimen and tested in the same manner.

4.2.4.5 Nitrogen compounds. Dissolve 1.000 g of the specimen in 40 ml of water in a distillation flask and add 1 g of powdered Devarda metal and 10 ml of 10-percent sodium hydroxide solution. Allow to stand for 2 hours protected from gain or loss of ammonia. Distill 40 ml of the solution, collecting the distillate in 5 ml of water containing 1 drop of dilute hydrochloric acid. To the distillate solution, add 1 ml of 10-percent sodium hydroxide solution and 2 ml of Nessler solution. Any color produced shall be no darker than that produced when 0.04 mg of ammonium chloride is substituted for the specimen and tested in the same manner.

4.2.4.6 Sulfate. Dissolve 3.00 g of the specimen in 50 ml of water. Add 1 ml of 10-percent hydrochloric acid and 5 ml of 12-percent barium chloride solution and allow to stand overnight. No turbidity or precipitate shall be formed in the solution.

4.2.4.7 Calcium, magnesium, and ammonium hydroxide precipitate. Weigh to the nearest 0.1 g approximately 10 g of the specimen. Boil with 50 ml of water and 15 ml of hydrochloric acid until chlorine evolution has stopped. Dilute with water to 120 ml, heat to boiling, and add 5 ml of 4-percent ammonium oxalate solution, 3 ml of 13-percent dibasic ammonium phosphate solution, and 20 ml of 28-percent ammonium hydroxide solution. Allow to stand overnight. Filter any precipitate through a tared filter crucible weighed to the nearest 0.1 mg. Wash with 2.5-percent ammonium hydroxide solution and ignite to constant weight. Cool to room temperature and weigh to the nearest 0.1 mg. Calculate the percent calcium, magnesium, and ammonium hydroxide preciptate as follows:

$$\text{Percent calcium, magnesium, and ammonium hydroxide precipitate} = \frac{100(A-B)}{W}$$

where: A = Weight of filter crucible and residue in grams,
B = Weight of filter crucible in grams, and
W = Weight of specimen in grams.

4.2.4.8 Heavy metals. To 3.000 g of the specimen, add 10 ml of water and 10 ml of hydrochloric acid (this is solution A). To 1.000 g of the specimen, add 10 ml of water, 10 ml of hydrochloric acid, and 0.02 mg of lead ion (this is solution B). Evaporate both solutions to dryness on a steam bath. To each residue, add 5 ml of hydrochloric acid and

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re-evaporate to dryness. Take up each residue in 25 ml of water and to each solution add 1 drop of phenolphthalein indicator and neutralize with 0.1N sodium hydroxide solution. Add 1 ml of 1N acetic acid to each solution, dilute each solution to 40 ml with water, and add 10 ml of a freshly-prepared saturated aqueous solution of hydrogen sulfide to each solution. Any color produced in the solution A test shall be no darker than that produced in the solution B test.

4.2.4.9 Iron. To 2.000 g of the specimen, add 10 ml of water and 10 ml of hydrochloric acid. Evaporate to dryness on a steam bath. Add 5 ml of hydrochloric acid and re-evaporate to dryness. Prepare a control by adding 15 ml of hydrochloric acid to 0.01 mg of iron and evaporating on a steam bath to dryness. Take up the specimen solution and the control separately in 2 ml of hydrochloric acid and water to make 50 ml. To each solution add 30 mg of ammonium persulfate and 3 ml of 25-percent ammonium thiocyanate solution. Any red color produced in the specimen solution shall be no darker than that produced in the control.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound ± 0.1 ounce of sodium chlorate shall be packaged in a type 1, class 1 bottle utilizing inclosure b (bottles in metal cans) in accordance with the requirements for level A packaging as specified in PPP-C-30[illegible].

5.1.1.1 Intermediate packaging. Six 1-pounds bottles of sodium chlorate, packaged as specified in 5.1.1, shall be intermediately packaged upright in a three bottle by two bottle pattern in a snug-fitting box conforming to style RSC, grade W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements for class weather-resistant boxes, by taping, as specified in the appendix to PPP-B-636.

5.1.2 Level C. Quantities of sodium chlorate as specified (see 6.2) shall be packaged to provide adequate protection against contamination, deterioration, and damage from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Eight boxes of sodium chlorate, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The box shall be closed and strapped using galvanized strapping in accordance with the applicable requirements in the appendix to PPP-B-621.

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5.2.2 Level B. Sodium chlorate shall be packed as specified in 5.2.1, except that a fiberboard box shall be used instead of the nailed wood box. The fiberboard box shall conform to style RSC, grade V3c of PPP-B-636 and shall be closed by taping in accordance with the requirements for taping RSC boxes to be shipped as exterior containers, and then reinforced by tape banding as specified in the appendix to PPP-B-636.

5.2.3 Level C. Sodium chlorate, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Department of Transporation regulations or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers and shipping containers specified in 5.3.1 and any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special marking.

5.3.1.1 Unit containers. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Precautionary information:

WARNING! STRONG OXIDANT

CONTACT WITH OTHER MATERIAL MAY CAUSE FIRE

Clothing and vegetation contaminated with sodium chlorate or its solution are DANGEROUSLY FLAMMABLE. Remove clothing and wash thoroughly with water.
Spillage may cause fires. Do not get on floor. Sweep up and remove immediately.
When not in use, keep tightly closed in original container.
Keep away from fire.
Store separately from flammable material.

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5.3.1.2 Shipping containers. Each shipping container shall be marked in accordance with Department of Transportation regulations to show "OXIDIZING MATERIAL"; in addition, the yellow Department of Transportation label shall be applied.

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the hardwood, 4-way, 40- by 48-inch, double-wing pallet of MIL-P-15011 except that the strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness (see 6.2).

5.5 Wood preservation. When specified in the contract or order, wood shipping containers, pallets, and storage aids shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to type II, composition A of TT-W-572 (see 6.2).

6. NOTES

6.1 Intended use. Sodium chlorate is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity required for level C packaging.
(d) Special marking other than 5.3.1, if required.
(e) Palletization, if required.
(f) Wood preservation, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

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6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-463, dated 18 April 1968.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0810

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. *Fold on lines on reverse side, staple in corner, and send to preparing activity.* Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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